

Design and Impact Analysis of Hybrid Foam-Filled Voronoi Sandwich Panels for Vehicle Crumple Zones

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Abstract

In order for a car to be crash-worthy, its components must be light in weight and efficient at absorbing the impact of a crash. In a conventional sandwich panel with a uniform honeycomb core, high peak impact forces and irregular collapse modes are often observed under dynamic axial loading, thus limiting their use in car crumple zones. To overcome these limitations, this paper proposes the design and impact response analysis of a hybrid foam-filled Voronoi sandwich panel, which is modeled after natural cellular structures. Nine different Voronoi sheet layouts were systematically developed using the Taguchi design approach & Rhino Software and analyzed using static structural analysis to determine an optimal cellular design. With the optimized design, suitable materials for the Voronoi sheets and foam core were chosen, and a hybrid multi-layer sandwich panel was designed. Explicit dynamic finite element analysis was carried out to analyze the impact response of the proposed panel under axial loading conditions simulating car crumple zones. The analysis results show that the foam-filled Voronoi sandwich panel has superior progressive collapse characteristics, lower peak impact force, and better energy absorption capacity than the conventional uniform-core sandwich panel. These results demonstrate the feasibility of hybrid Voronoi-foam sandwich panels as efficient energy-absorbing structures in car passive safety systems.

Keywords: Mechanical engineering, Automotive, bio-inspired structures, crumple zone, crashworthiness, modeling and simulation.

1. Introduction

One of the significant challenges facing contemporary automotive engineering is the enhancement of vehicle crashworthiness while simultaneously preserving a lightweight structural design. Structural components within automobiles must possess the capability to absorb substantial impact energy without incurring excessive mass, particularly as there is an increasing focus on passenger safety, fuel efficiency, and emissions reduction. The crumple zone, typically situated at the front and rear sections of a vehicle, serves a vital function in dissipating collision energy through controlled deformation, thereby minimizing the stresses transmitted to the passenger compartment. As a result, the development of lightweight energy-absorbing structures has emerged as a crucial area of research within automotive safety engineering.

Sandwich structures are extensively employed in lightweight engineering applications due to their favorable stiffness-to-weight ratio and superior structural efficiency. Conventional sandwich panels typically comprise two face sheets bonded to a lightweight core material, such as foam or honeycomb. Honeycomb cores, while characterized by low density and commendable compressive strength, often display anisotropic behavior, elevated initial peak forces, and unstable collapse mechanisms when subjected to dynamic axial loading conditions. These attributes may result in suboptimal crash energy management in automotive structures, leading to abrupt force spikes and non-uniform deformation patterns.

In response to these limitations, bio-inspired cellular structures have garnered increasing attention in recent years. Among these, Voronoi cellular architectures, derived from natural tessellation patterns, feature irregular cell geometries that facilitate more uniform stress distribution and enhanced progressive collapse behavior under impact loading. Compared to traditional periodic honeycomb structures, Voronoi configurations diminish the likelihood of synchronized cell wall buckling, thereby improving energy absorption performance. Nonetheless, standalone cellular structures may still encounter premature densification or localized instability under extreme impact loads, which constrains their overall crashworthiness.

Recent investigations have indicated that the integration of cellular structures with foam cores can significantly bolster impact resistance and energy absorption capacity. Foam-filled lattice structures and sandwich panels have demonstrated enhanced structural stability and diminished peak crushing forces, attributable to the interaction between the cellular framework and the compressible foam material. The foam layer provides additional energy absorption

through progressive compression and densification, thereby improving the overall crashworthiness of the structure.

In this context, the present study proposes the design and impact analysis of a hybrid foam-filled Voronoi sandwich panel, aimed at localized energy absorption in automotive crumple zones. A structured design approach, based on the Taguchi Design of Experiments (DOE) method, is employed to generate nine distinct Voronoi sheet configurations by varying key geometric parameters. Static structural analysis is conducted to evaluate these configurations and identify an optimal cellular design exhibiting favorable mechanical performance. Subsequently, explicit dynamic finite element simulations are carried out to assess the crash performance of the optimized hybrid sandwich structure under axial impact loading conditions.

The organized combination of geometric optimization and dynamic validation to develop a hybrid Voronoi–foam sandwich structure with improved energy absorption capabilities is the primary contribution of this research. This study seeks to illuminate the process of designing advanced cellular sandwich panels for application in passive safety systems within vehicles.

2. Design of Voronoi Sandwich Panel

The hybrid energy-absorbing structure examined in this research is a sandwich panel made up of two Voronoi cellular sheets that are separated by a foam core. The overall thickness of the core in the sandwich structure measures 22 mm, comprising two Voronoi layers, each with a thickness of 6 mm, along with a central foam layer that is 10 mm thick. The sandwich panel has a square planform with dimensions of 200 mm $\times$ 200 mm.

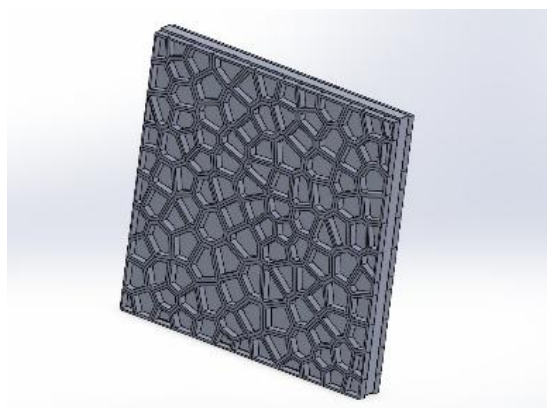


Figure 1 Voronoi Sandwich Panel

The Voronoi sheets serve as the main load-bearing elements of the sandwich structure and trigger progressive deformation when subjected to compressive loads. These layers effectively absorb a considerable amount of impact energy through the bending and plastic deformation of the lattice struts. The foam core, situated between the Voronoi sheets, enhances the overall crashworthiness of the structure by providing additional energy absorption through the compression and densification of the cellular material.

2.1 Design of Voronoi sheet

The Voronoi cellular structures were created using stochastic seed distributions within the design domain to produce irregular polyhedral cell geometries with the help of the rhino software & grasshopper add-on. This stochastic topology enhances load redistribution during deformation and reduces the chances of localized stress concentration, thus facilitating stable progressive collapse during impact loading.

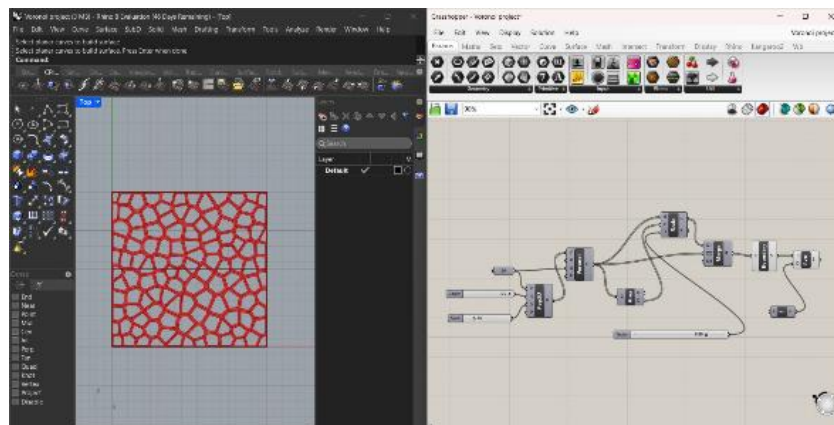


Figure 2 Voronoi Sheet formation using Rhino & Grasshopper

2.1.1 Taguchi Design of Experiment

To methodically examine how Voronoi geometric variables affect the crashworthiness behavior of the sandwich structure, the Taguchi design of experiments (DOE) technique was used. The Taguchi method offers a practical statistical strategy for studying the impact of several design factors while greatly lowering the number of simulations needed compared with a traditional full factorial plan. As a result, parameter interactions can be investigated and the best design settings can be identified with limited computational cost.

In this work, three primary geometry-related factors defining the Voronoi cellular architecture were chosen for optimization: cell count, seed distribution, and strut thickness factor. Each factor was evaluated at three separate levels to represent changes in the Voronoi

lattice geometry and to observe how these changes influence the structural response. The cell count determines the tessellation density of the Voronoi pattern, and therefore affects how many load-carrying members exist in the structure. The seed distribution factor controls the degree of spatial randomness and the resulting cell topology, which in turn alters deformation modes and the way stresses are distributed. The strut thickness factor specifies the relative thickness of the lattice struts and has a direct effect on stiffness and overall load-supporting capacity within the cellular framework.

Using these three factors and three levels, a Taguchi L9 orthogonal array was selected to plan the numerical trials. This array cuts the simulation requirement from 27 cases in a full factorial design down to nine simulations, while preserving orthogonality and balanced representation of the factors. For every run, finite element analyses were performed to extract the maximum reaction force and the maximum directional deformation of the sandwich structure under the applied loading scenario.

To compare the robustness and performance of each parameter combination, the signal-to-noise (S/N) ratio was computed using the larger-the-better formulation, because greater peak force corresponds to stronger crashworthiness and improved load-carrying performance. The S/N metric captures both average performance and scatter, enabling the selection of parameter levels that raise structural strength while reducing sensitivity to variations. ANOVA shows strut thickness is the most influential factor on maximum force, with the highest F-value and a very low p-value confirming strong statistical significance. Cell count also meaningfully affects load-bearing capacity by changing Voronoi lattice density, while seed distribution has a smaller impact despite influencing geometric randomness.

Larger-the-better: $S/N=20\log_{10}(\text{Max Force Reaction})$

Cell count	Seed	Strut Thickness Factor	Max Force Reaction (N)	Max Directional Deformation (m)	SN Ratio (dB)
120	10	0.8	38837	0.00037686	91.7849135
80	30	0.8	36419	0.031788	91.22656033
100	50	0.8	36412	0.030279	91.22489068
120	30	0.85	25123	0.0015714	88.00142997
100	10	0.85	24537	0.00068038	87.79642926
80	50	0.85	22153	0.00023111	86.90865095
120	50	0.9	12767	0.00078656	82.12177717
100	30	0.9	12725	0.001107	82.09315582
80	10	0.9	11910	0.03051	81.51823523

Table 1 Taguchi Implementation for the Voronoi sheets and the results obtained in Ansys

Source	Degrees of Freedom (DF)	Sum of Squares (SS)	Mean Square (MS)	F-Value	P-Value
Cell Count	2	0.872	0.436	27.05	0.0356
Seed	2	0.211	0.105	6.54	0.1326
Strut Thickness Factor	2	137.052	68.526	4251.21	0.0002
Error	2	0.032	0.016	—	—
Total	8	138.167	—	—	—

Table 2: ANOVA table

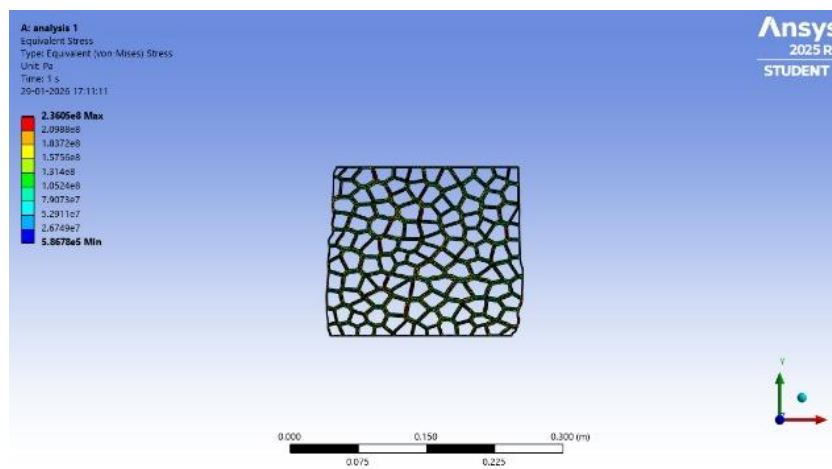


Figure 3 Static Structural Analysis

Here, The Voronoi sheet with a cell count 120, seed of 10 and strut thickness factor of 0.8 is used in the project due to its highest force reaction and lowest deformation relative to the load.

2.2 Design of the foam

The foam core was engineered to emulate a closed-cell cellular material, frequently utilized in structural sandwich panels due to its exceptional compressive strength, shear stiffness, and structural stability.

Closed-cell foams are characterized by isolated pores encased by slender solid cell walls, enabling them to endure compressive loads while experiencing gradual densification during deformation.

In order to simulate the energy-absorbing behavior of cellular structures, the foam core used in the sandwich panel was chosen based on its material properties. The material was depicted using its mechanical characteristics, such as density, elastic modulus, and plastic

hardening behavior, rather than explicitly modeling the intricate internal geometry of the foam. This method preserves computational efficiency during the explicit dynamic analysis while enabling the simulation to represent the foam's compressive and energy-absorbing properties.

3. Material Selection

The material chosen for the Voronoi cellular sheet is Aluminium alloy Al 5052-H32, which is known for its advantageous blend of strength, ductility, and resistance to corrosion, rendering it appropriate for structural applications in the automotive sector.

Aluminium 5052	
Property	Value
Density (ρ)	2680 kg/m ³
Young's Modulus (E)	7×10^{10} Pa
Poisson's Ratio (ν)	0.33
Bulk Modulus (K)	6.86×10^{10} Pa
Shear Modulus (G)	2.63×10^{10} Pa
Yield Strength (σ_y)	1.93×10^8 Pa
Tangent Modulus(E_t)	1.5×10^8 Pa

Table 3 Material Property of Al-5052 used

The material used for the core has been identified as aluminium foam on the basis of its mechanical properties and the way it deforms under compression. Foam materials are characterised by their low density, low elasticity and high energy absorption capacity, which distinguish them from solid metals.

Property	Value
Density (ρ)	400 kg/m ³
Young's Modulus (E)	5×10^8 Pa
Poisson's Ratio (ν)	0.30
Bulk Modulus (K)	4.17×10^8 Pa
Shear Modulus (G)	1.92×10^8 Pa
Plastic Hardening (Multilinear)	
Stress (Pa)	Plastic Strain
3.0×10^6	0
3.2×10^6	0.05
3.5×10^6	0.10
4.0×10^6	0.20
6.0×10^6	0.40
8.0×10^6	0.50

Table 4 Material Property for Aluminium foam used

The basic material was defined in the simulation model by its typical characteristics of aluminium foamed material, such as its low density and non-linear plastic deformation. These characteristics enable the material to undergo a gradual plastic collapse when subjected to compression or impact loads, which is the defining characteristic of metallic foams. The overall mass of the sandwich panel after giving the material properties to the panel and foam is 0.62311 kg

4.Simulation Test

Finite element analysis was performed to assess the structural behavior of the hybrid sandwich panel when subjected to compressive loading conditions. Three-dimensional models of the Voronoi cellular sheets and foam core were created using Solidworks software and subsequently imported into the Ansys Explicit and Abaqus for simulation. An impact speed of 20 m/s (about 72 km/hr) was selected to replicate a realistic vehicle crash condition commonly applied in crashworthiness studies. This speed enables evaluation of the structure's energy absorption capacity and deformation response when subjected to intense dynamic impact loading.

4.1 Drop Ball Impact Analysis

Drop ball analysis simulates high-velocity impact circumstances and evaluates the structure's deformation, stress distribution, and energy absorption capabilities under dynamic loading. It determines if the structure can sustain impact and effectively dissipate energy.

A drop ball impact simulation was used to examine the sandwich panel's surface impact performance. In this instance, the Voronoi sandwich structure's surface experienced impact loading due to the impactor's initial velocity of 20 m/s acting parallel to the sandwich panel's face.

Similarly, another explicit dynamic drop ball analysis was used to assess the Voronoi sandwich panel's axial impact performance. In this simulation, the Voronoi core region was impacted by a spherical impactor that had the same starting velocity of 20 m/s and was pointed perpendicular to the sandwich panel's axis. The plate beneath the panel is given as fixed support in both cases

Parameter	Description
Mass of the Impactor	110.87 kg
Element Shape	Tetrahedral
Element Size in Sandwich Structure	5 mm
Total Number of Elements	23,558
Global Element size	21.556 mm

Table 5 Mesh Details for Ansys Simulation

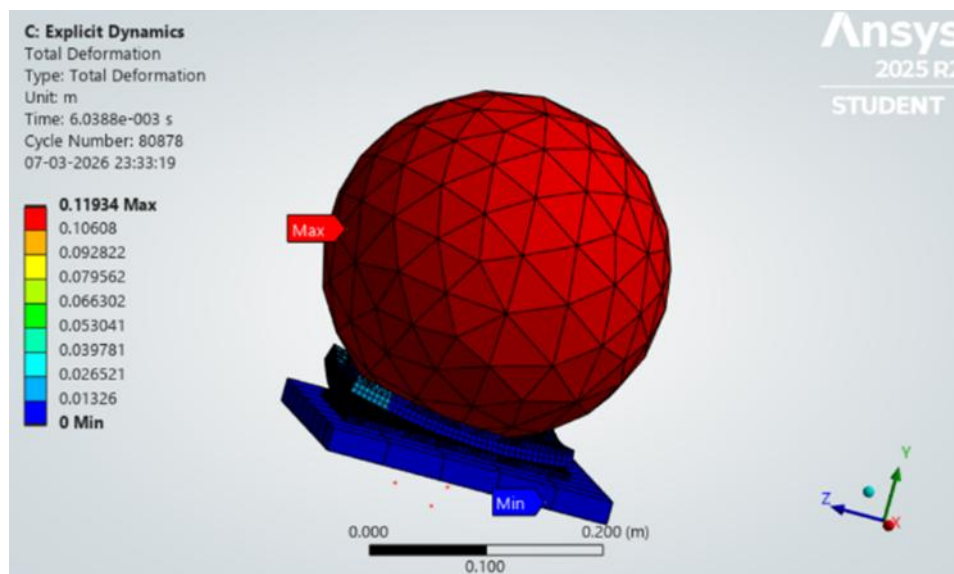


Figure 4 Drop Ball Impact Analysis (on surface)

4.2 Plate crush Analysis

The plate crush analysis was conducted to assess the compressive behavior and energy absorption capacity of the Voronoi sandwich panel when exposed to crushing loads. This analysis is crucial for understanding the structural response to progressive compressive forces, which is important for evaluating the crashworthiness and structural integrity of sandwich panels.

The rigid plate in the axial crushing configuration was placed perpendicular to the sandwich panel and moved in the panel's normal direction, creating compressive loading across

the structure's thickness. The imposed stress was first transferred to the upper face sheet during the crushing process, and it then moved through the foam layer and Voronoi lattice core.

Parameter	Description
Element Type	C3D8R
Element Shape	Hexahedral elements
Global Element Size	3 mm
Total Number of Elements	~30,000
Mesh Convergence Check	Performed to ensure mesh-independent results

Table 6 Mesh Details for Abaqus Simulation

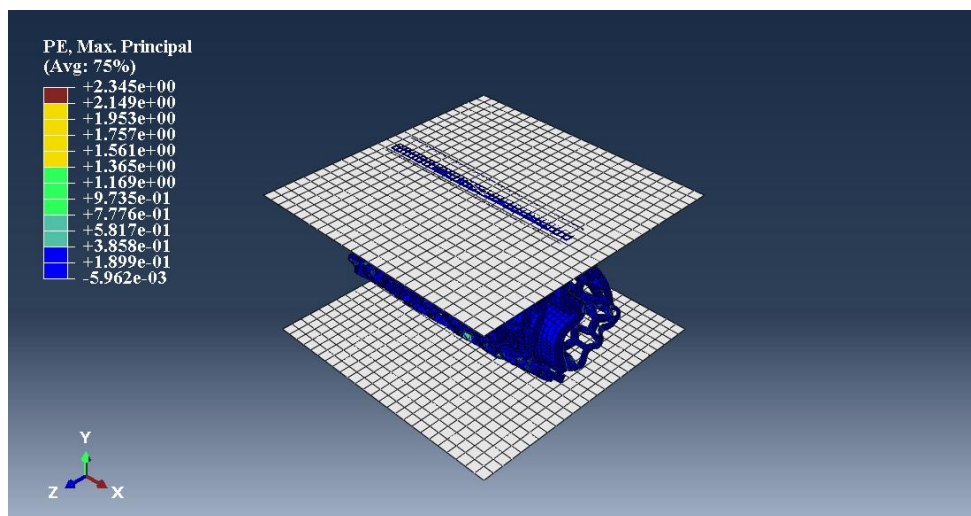


Figure 5 Plate Crush Analysis

5. Parameters Obtained

The key crashworthiness parameters considered in this project are widely used to evaluate the energy absorption capability and structural performance of sandwich panels under impact and crushing conditions. The associated parameters are given below:

(a) Energy Absorption

Energy Absorption (EA) refers to the total mechanical energy a structure can take in while it deforms under impact or crushing-type loads. In practical terms, it reflects the structure's capacity to dissipate incoming kinetic energy, primarily through plastic deformation rather than elastic rebound. A higher EA generally indicates better crashworthiness because more energy is consumed in controlled deformation. Mathematically, energy absorption is obtained by integrating the crushing force over the deformation distance:

$$EA = \int F \cdot dx$$

(b) Specific Energy Absorption (SEA)

Specific Energy Absorption describes how much energy is absorbed per unit mass of the structure. Because weight is critical in many applications (such as automotive, aerospace, and protective systems), SEA is widely used to compare the energy-absorbing efficiency of different materials and designs on an equal-mass basis. A structure with high SEA can provide strong impact protection while remaining lightweight:

$$SEA = EA / m$$

(c) Peak Crushing Force (PCF)

Peak Crushing Force is the highest force recorded during crushing or impact. This value is important because excessive peaks can lead to sudden load transfer, increasing the risk of damage or injury in real-world crash scenarios. PCF is typically taken as the maximum force on the force–displacement curve:

$$PCF = F_{max}$$

(d) Mean Crushing Force (MCF)

Mean Crushing Force represents the average force sustained during progressive crushing over a given crush distance. It provides a useful measure of the overall load level maintained while absorbing energy and is often used to assess the stability of the crushing process:

$$MCF = EA / \delta$$

(e) Crushing Force Efficiency (CFE)

Crushing Force Efficiency indicates how uniformly the structure crushes by comparing the average crushing force to the peak value. A higher CFE suggests a smoother, more stable response with fewer extreme force spikes, which is typically desirable for safer and more predictable energy absorption:

$$CFE = MCF / PCF$$

6. Results and discussion

6.1 Drop Ball analysis

The significant results were obtained during the analysis in Ansys Software by giving the end time of 0.015 seconds.

Parameter	Drop Ball	
	Surface Impact	Axial Impact
Energy Absorption (J)	3812.2	6521.6
Specific Energy Absorption (J/kg)	6116.6	10464.7
Peak Crushing Force (N)	38900	96900
Mean Crushing Force (N)	1.008×10^6	1.647×10^5
Crushing Force Efficiency (CFE)	0.0386	0.583

Table 7 Crashworthiness values of Drop Ball Analysis

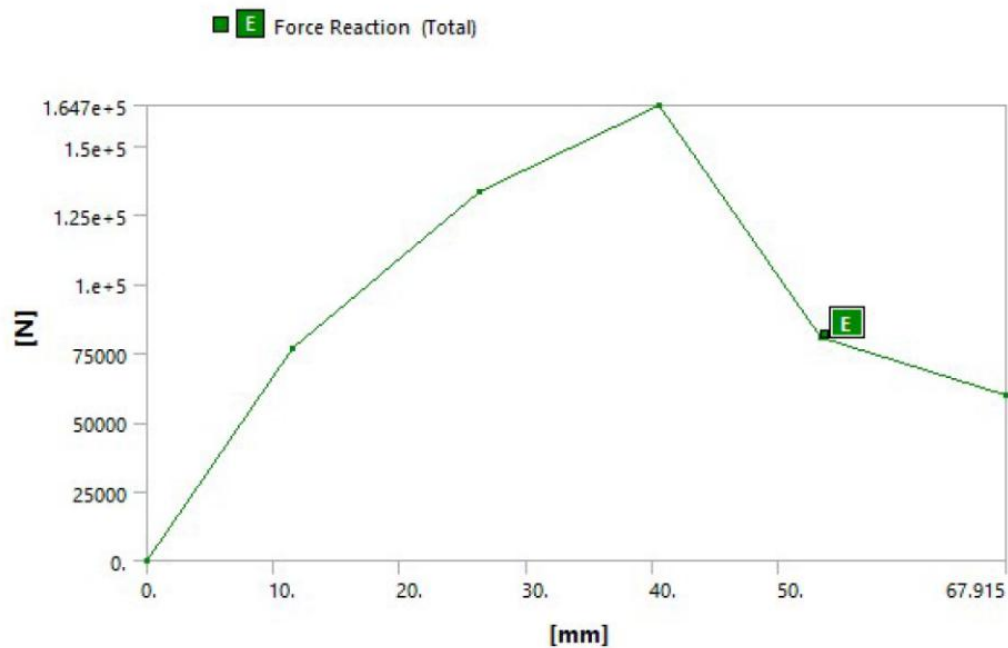


Figure 6 Force-Displacement Graph for drop ball test(axial)

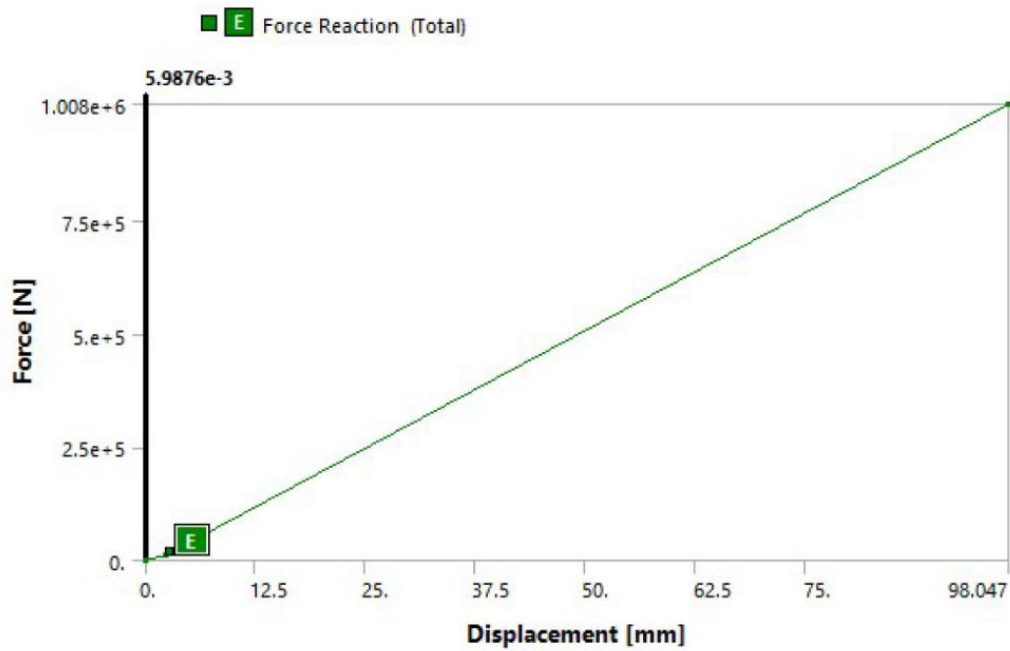


Figure 7 Force-Displacement Graph for drop ball test (surface)

6.2 Plate analysis

The significant results were obtained during the analysis in Abaqus Software by giving the end time of 0.015 seconds.

Parameters	Values
Energy Absorption (J)	15,788.2
Specific Energy Absorption (J/kg)	25,337.74
Peak Crushing Force (N)	229,314
Mean Crushing Force (N)	83,219.7
Crushing Force Efficiency (CFE)	0.363

Table 8 Crashworthiness values of Plate Crush Analysis

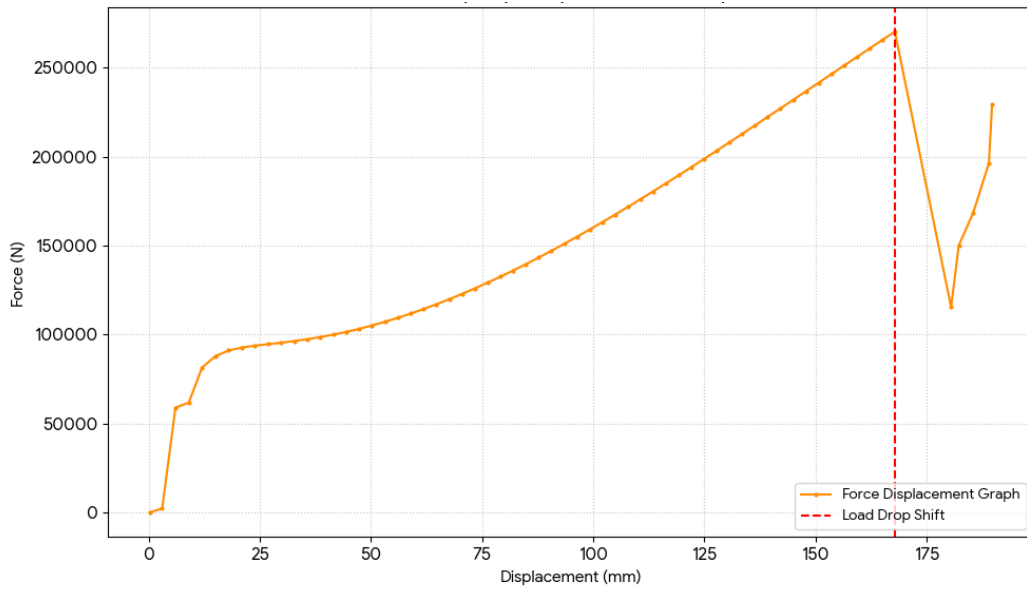


Figure 9 Force-Displacement Graph for plate crush

7. Conclusion

The present study shows that combining Voronoi-based cellular face sheets with a foam core, and then refining the design through the systematic use of the Taguchi optimisation method, offers a practical, accurate and computationally efficient route to improving the crashworthiness of sandwich panel structures. Rather than relying on lengthy trial-and-error iterations, the Taguchi-based design of experiments provides a structured framework for screening design variables, quantifying their influence, and quickly identifying high-performing configurations. Within this framework, the work clearly highlights which geometric parameters most strongly govern performance: strut thickness was found to be the dominant factor, with cell density and seed distribution also playing meaningful roles in shaping the overall deformation and energy absorption response.

To build a robust picture of crashworthiness, the hybrid foam-filled Voronoi sandwich panel was evaluated in detail through numerical simulations carried out in Ansys Explicit and Abaqus across multiple loading cases. These scenarios were chosen to represent a range of realistic impact and crushing conditions and included axial drop-ball impact, surface drop-ball impact, and quasi-static plate crushing. Taken together, these simulation programmes enabled a thorough assessment of deformation modes, progressive collapse behaviour, energy absorption capacity, and the way loads are transferred and redistributed through both the cellular face sheets and the foam core during crushing.

Results from the drop-ball impact studies show a clear difference between axial and surface impact responses. Under axial impact conditions, the structure absorbed markedly more energy than under surface impact, although this benefit was accompanied by higher peak forces. This trade-off is important for crash design, as it suggests the hybrid panel is highly effective at dissipating energy when the impact direction promotes more global engagement of the cellular network and foam core. The higher energy absorption is consistent with extensive plastic deformation of the struts and a more pronounced progressive collapse sequence, where crushing propagates through the structure rather than being limited to a small region. In other words, axial loading encourages a broader portion of the panel to participate in energy dissipation, increasing the amount of work done through plastic folding, bending and crushing.

By comparison, surface drop-ball impact led to lower overall energy absorption and reduced crushing efficiency, pointing to a more localised deformation response. In surface loading, the impact tends to concentrate stresses near the contact zone, which can limit how effectively the rest of the structure contributes to dissipating energy. This localised behaviour can reduce the available collapse length and therefore restrict the amount of stable progressive crushing that is typically desired in energy-absorbing components. The simulations therefore underline that, while the hybrid concept performs well across conditions, the direction and nature of loading have a strong influence on whether the deformation is distributed across the panel or concentrated around the impact site.

The quasi-static plate crushing simulations further reinforced the stability advantages of the hybrid configuration. The panel demonstrated smooth, progressive deformation with improved crushing force efficiency, indicating that it can maintain controlled collapse rather than exhibiting irregular or unstable failure. The force–displacement curves showed a consistent load transfer mechanism and reduced fluctuations in crushing force, characteristics that are widely regarded as favourable for crash energy management. A more stable force response generally implies less risk of sudden load spikes, which can be problematic in protective structures, and indicates that the geometry and foam filling work together to promote steady collapse.

Across all loading cases, the study confirms that the hybrid foam-filled Voronoi sandwich panel achieves superior crashworthiness compared with conventional cellular designs. The Voronoi geometry helps distribute stresses more evenly and provides multiple load paths, reducing the likelihood that a single weak region dominates failure. Meanwhile, the foam core contributes additional energy dissipation and supports the cellular face sheets, which can delay local buckling, stabilise strut deformation, and improve the overall progressive

collapse response. This combination leads to controlled deformation behaviour, enhanced energy absorption, and a tendency towards stable crushing rather than brittle or abrupt failure modes.

From an application standpoint, these findings point to strong potential for implementing such hybrid structures in automotive crash components, particularly in crumple zones where efficient energy dissipation, predictable deformation, and passenger safety are paramount. The ability to tailor performance through key geometric parameters—especially strut thickness—also offers practical design flexibility, allowing engineers to target specific force levels and energy absorption requirements for different vehicle architectures or crash scenarios. Looking ahead, future work could focus on extending the optimisation to include geometric gradients (for example, varying cell size or strut thickness spatially), exploring alternative material pairings for the face sheets and foam, and conducting experimental validation to confirm real-world behaviour under dynamic crash conditions. Such steps would help bridge the gap between simulation-led design and deployable, manufacturable solutions that can deliver consistent performance in service.

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